One-Shot Flow, Any-Time Frame: A Bidirectional Warping Framework for Event-Based Video Frame Interpolation

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Abstract

Video Frame Interpolation (VFI) is a crucial task in video processing. Flow-based methods, despite their success, are constrained by a fundamental dilemma: forward warping is efficient but prone to artifacts, while backward warping yields higher quality at a significant computational cost, especially for multi-frame interpolation. This trade-off is a major bottleneck. To overcome this, we introduce "One-Shot Flow, Any-Time Frame," a novel framework for Event-based VFI (E-VFI) that achieves both high efficiency and superior quality for arbitrary-time interpolation. Our framework uniquely computes a comprehensive motion trajectory representation in a single pass using a Bidirectional Flow Estimation Block (BiFEB), leveraging the high temporal resolution of event data. Subsequently, our Flow Query (FQ) module can instantly retrieve the bidirectional optical flow for any timestamp, enabling the generation of any number of frames without repeated computation. Finally, a novel Bidirectional Warping (BiW) mechanism intelligently fuses the strengths of both warping directions, effectively mitigating artifacts and producing high-fidelity results. Extensive experiments show that our approach consistently surpasses state-of-the-art E-VFI methods in both reconstruction quality and inference efficiency, representing a substantial advance in efficient and high-quality event-based video interpolation. The project page is available at: https://github.com/Sudadaaaa/OF-AF

1. Introduction

Video Frame Interpolation (VFI) is a fundamental task in computer vision with widespread applications such as slow-motion video [1, 13, 42], video compression [40], and view synthesis [48]. Among various VFI techniques, optical flow-based methods have become the dominant paradigm due to their ability to generate sharp and geometrically consistent results [9, 11, 15, 16, 22]. These methods fall into two categories, each with its own limitations.

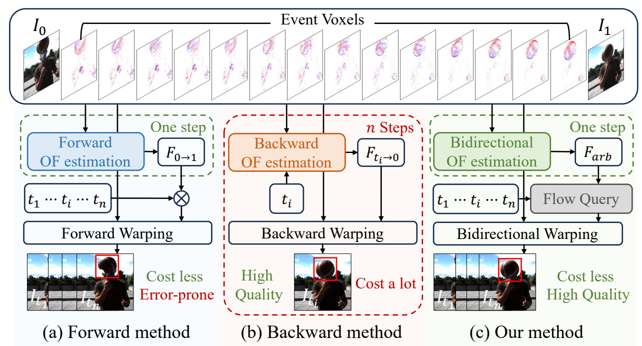

Figure 1. Comparison of different optical flow methods. (a) Forward method enables fast interpolation but produces artifacts such as holes. (b) Backward method improves quality via backward warping but requires heavy computation. (c) Our method unifies both advantages, achieving high-quality synthesis with low cost.

Forward methods achieve high efficiency for a large number of frames by estimating optical flow ($I_0 \rightarrow I_1$) only once and scaling it linearly for t (Fig. 1(a)). However, this linear approximation struggles with complex motion, and forward warping often leaves regions uncovered, producing holes and degraded image quality (Fig. 2(a)). In contrast, backward methods ensure full pixel coverage and superior quality by sampling every target pixel from the source frames via reverse optical flow ($I_1 \rightarrow I_0$) (Fig. 2(b)). However, predicting flow for every interpolated frame incurs substantial computational costs, creating a critical trade-off between computational cost and quality (Fig. 1(b)).

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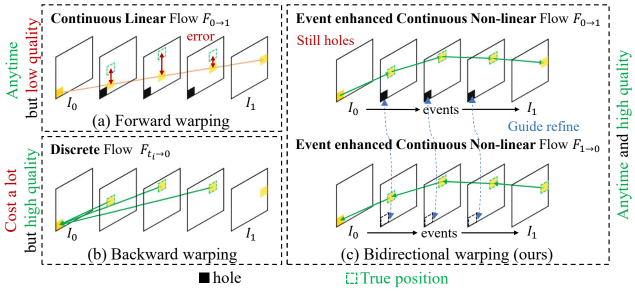

Figure 2. Comparison of different warping. Our method combines the advantages of forward warping and backward warping.

The advent of event cameras [38] has offered a promising direction for breaking this trade-off. By capturing per-pixel brightness changes with microsecond-level precision, event data provides dense, continuous motion cues that complement conventional VFI inputs, leading to notable progress in Event-based VFI (E-VFI). However, existing E-VFI methods [18, 25, 29, 38, 39] still fail to overcome the fundamental efficiency–quality dilemma. To alleviate the costly per-frame estimation inherent to their backward warping foundations, several works [25, 29] adopt iterative strategies, which improve inference efficiency but introduce substantial memory overhead that grows rapidly with the number of interpolated frames. Ultimately, constrained by the backward warping paradigm itself, these methods remain unable to achieve true efficiency–quality unification.

To break this impasse, we propose a new framework, “One-Shot Flow, Any-Time Frame,” which synergistically fuses the advantages of forward warping, backward warping, and event data. Our key insights are as follows: *(1) The linear motion assumption inherent to forward flow causes accumulated errors, but dense event streams provide continuous, non-linear motion cues that can correct it; and (2) the hole artifacts produced by forward warping can be effectively filled by leveraging backward flow information, ensuring reconstruction quality.* Building on these insights, our framework performs one-shot motion estimation and enables arbitrary-time interpolation via efficient flow querying and adaptive refinement (Fig. 1(c)). At its core, the Bidirectional Flow Estimation Block (BiFEB) jointly processes keyframes and event streams to build a unified motion representation. The Flow Query (FQ) module then retrieves accurate flows for any intermediate timestamp, while the Bidirectional Warping (BiW) module detects unreliable regions and guides refinement to produce high-quality, artifact-free frames. By harmonizing these components, our method unifies the efficiency of forward approaches with the fidelity of backward ones, marking a significant step toward bridging the efficiency–quality gap in VFI. The main contributions of this work are summarized as follows:

- We propose a novel event-based video interpolation framework that captures continuous motion trajectories in a single forward pass, enabling the rapid and memory-efficient synthesis of a large number of high-quality intermediate frames at arbitrary timestamps.
- We introduce the Bidirectional Flow Estimation Block (BiFEB), which estimates a holistic representation of bidirectional optical flows, and a Flow Query (FQ) module that allows efficient flow retrieval for any target time.
- We design the Bidirectional Warping (BiW) module, which intelligently repairs warping artifacts by generating adaptive restoration masks and sourcing contextual information from reference frames.
- Extensive experiments on both synthetic and real-world datasets demonstrate that our method significantly outperforms state-of-the-art E-VFI approaches, achieving superior visual quality while substantially reducing computational and memory costs

2. Related Work

Video frame interpolation. Video Frame Interpolation (VFI) has historically been pursued via two main paradigms: synthesis-based and flow-based methods. Synthesis-based approaches, including kernel-based [3, 4, 17, 21, 32, 33, 35] and diffusion models [12, 47], generate pixels directly. However, since they lack strong geometric constraints, they often result in blurry or temporally inconsistent outputs, especially in scenes with complex motion. In contrast, flow-based methods [9, 13, 20, 31] have emerged as the dominant paradigm. By leveraging explicit motion estimation via optical flow, they warp pixels from existing frames to synthesize intermediate ones, producing significantly sharper and more coherent results. However, their performance is fundamentally limited by the sparse temporal information between keyframes. This makes estimating complex, non-linear motion an ill-posed problem and creates a critical information bottleneck that hinders quality.

Event-based video frame interpolation. Event cameras provide dense motion cues that alleviate the temporal information bottleneck in conventional VFI, driving rapid progress in Event-based VFI (E-VFI) [6, 7, 18, 23, 25–29, 36, 38, 39, 41, 43, 46]. Existing methods fall into two categories: flow-free and flow-based approaches.

Flow-free methods [2, 6, 26, 28] directly synthesize pixels from events, bypassing motion estimation. However, by forgoing explicit motion modeling, they often struggle to preserve fine-grained textures and are prone to artifacts, as the synthesis process lacks robust geometric guidance.

Flow-based methods [27, 39, 41] exploit events to improve optical flow estimation. Recent works follow two strategies: 1) Single-frame estimation [18, 38], which predicts a precise flow for each target timestamp. This yields strong quality but incurs prohibitive computational cost when many frames are required. 2) Multi-frame estimation

[25, 29], which refines flows across multiple timestamps to reduce redundant computation. However, it still requires a flow prediction for every frame, and storing numerous intermediate variables leads to massive memory overhead.

In contrast, our method breaks this trade-off with a new paradigm: we compute a holistic, bidirectional motion representation for the entire time interval in a single pass, from which the optical flow at any timestamp can be retrieved on demand. This eliminates per-frame flow prediction and avoids the memory burden of iterative schemes, enabling both high efficiency and high-quality interpolation.

3. Proposed Method

3.1. Method Overview

Our proposed method takes the two keyframes (I_0 and I_1) and the intervening events as input and synthesizes an arbitrary number of intermediate frames. Fig. 3 shows the overall architecture, which is decomposed into two main stages: Bidirectional Flow Estimation and Bidirectional Warping.

Initially, the raw events are processed into a voxel grid representation, following [29, 38]. We then employ two separate feature extractors to derive $1/4$ scale features from both keyframes and the voxel grid. The Bidirectional Flow Estimation Block (BiFEB) then predicts the dense bidirectional optical flow between the two keyframes for arbitrary time t. Crucially, this step is performed only once, regardless of the number of frames subsequently interpolated, ensuring high computational efficiency.

Given a target time t, we rapidly obtain two sets of bidirectional optical flow ($F_{0\to t}, F_{t\to 0}$) and ($F_{1\to t}$, $F_{t\to 1}$) through Flow Querying. These flows are then utilized by the Bidirectional Warping (BiW) module to guide the network in synthesizing an accurate intermediate frame.

The following sections will detail the core components of our architecture, BiFEB and BiW.

3.2. Bidirectional Flow Estimation Block

To circumvent the need for repeatedly calculating optical flow for every time t, we introduce BiFEB. Similar to forward methods, BiFEB can estimate the bidirectional optical flow of objects at an arbitrary time between the two keyframes in a single pass. However, conventional forward methods typically assume linear and uniform motion between the keyframes, leading to inaccurate flow estimation when confronted with non-uniform or curvilinear motion. BiFEB addresses this limitation by using the high-temporal-resolution motion from events. Specifically, we divide the object's total motion over time into n continuous temporal segments, regardless of the target interpolation frame count (N). The events in each segment are then utilized to estimate the corresponding localized optical flow, enabling a more accurate optical flow between the two frames. This design is supported by a fundamental and reasonable motion assumption: object motion can be approximated as linear and uniform over extremely short intervals.

The architecture of BiFEB is detailed in Fig. 3. Focusing on the t-th temporal segment, BiFEB first aggregates the current events feature E_t with the contextual features θ_{t-1} and motion information V_{t-1} from the $(t-1)$-th segment to obtain the θ_t and V_t. Subsequently, we estimate the bidirectional optical flow by starting from the keyframe I_0. This process can be represented by Eq. (1).

$$\begin{aligned} \theta_t^0 &= RES(\theta_{t-1}^0, E_t), \\ V_t^0 &= RES(V_{t-1}^0, E_t, \theta_t^0), \\ F_{0\to t} &= FFE(V_t^0, \theta_t^0), \\ F_{t\to 0} &= BFE(V_t^0, \theta_t^0, D_{t-1}^0, F_{t-1\to 0}), \end{aligned} \tag{1}$$

where $(\cdot)^0$ denotes the variable processed from I_0. The Res denotes a residual block. The Forward Flow Estimator (FFE) module utilizes a GRU [37] to estimate the $F_{0\to t}$. The Backward Flow Estimator (BFE) estimates the backward flow $F_{t\to 0}$. *The detailed architecture of the BFE module is provided in the Supplementary Material.*

Initially, we set D_0^0 and θ_0^0 as the feature D_0, and set $F_{0\to 0}$ and V_0^0 as 0. We input the events in the reverse order and set D_0^1 and θ_0^1 as the feature D_1 to estimate the forward flow $F_{1\to t}$ and backward flow $F_{t\to 1}$. Finally, we obtain the bidirectional flow F_{arb} of the object at arbitrary time.

3.3. Flow Query

After BiFEB computes the F_{arb}, the Flow Query (FQ) module enables the efficient retrieval of bidirectional optical flows for any target timestamp $t \in (0, 1)$. We first query the bidirectional optical flow and the contextual information corresponding to the target time from the keyframe I_0. We begin by performing the querying using Eq. (2).

$$\begin{aligned} F_{0\to t_l}, F_{t_l\to t_r} &= Q(F_{arb}, 0\to t), \\ F_{t_r\to t_l}, F_{t_l\to 0} &= Q(F_{arb}, t\to 0), \\ \theta_{t_l}^0, \theta_{t_r}^0 &= Q(\theta, 0, t), \end{aligned} \tag{2}$$

where Q first determines the starting time $\mathbf{t}_l$ and ending time $\mathbf{t}_r$ of the time segment containing the input time t. Then, according to t_l and t_r, it queries F_{arb} and θ to obtain the bidirectional optical flow and contextual features at time t_l and t_r. Then, we use Eq. (3) to get the final output.

$$\begin{aligned} \lambda &= \frac{t - t_l}{t_r - t_l}, \\ F_{0\to t} &= F_{0\to t_l} + \lambda \cdot F_{t_l\to t_r}, \\ F_{t\to 0} &= (1-\lambda)\cdot F_{t_r\to t_l} + F_{t_l\to 0}, \\ \theta_t^0 &= (1-\lambda)\cdot \theta_{t_l}^0 + \lambda \cdot \theta_{t_r}^0, \end{aligned} \tag{3}$$

where the weight λ is utilized to interpolate the bidirectional optical flow and the contextual information at t_l and t_r. This

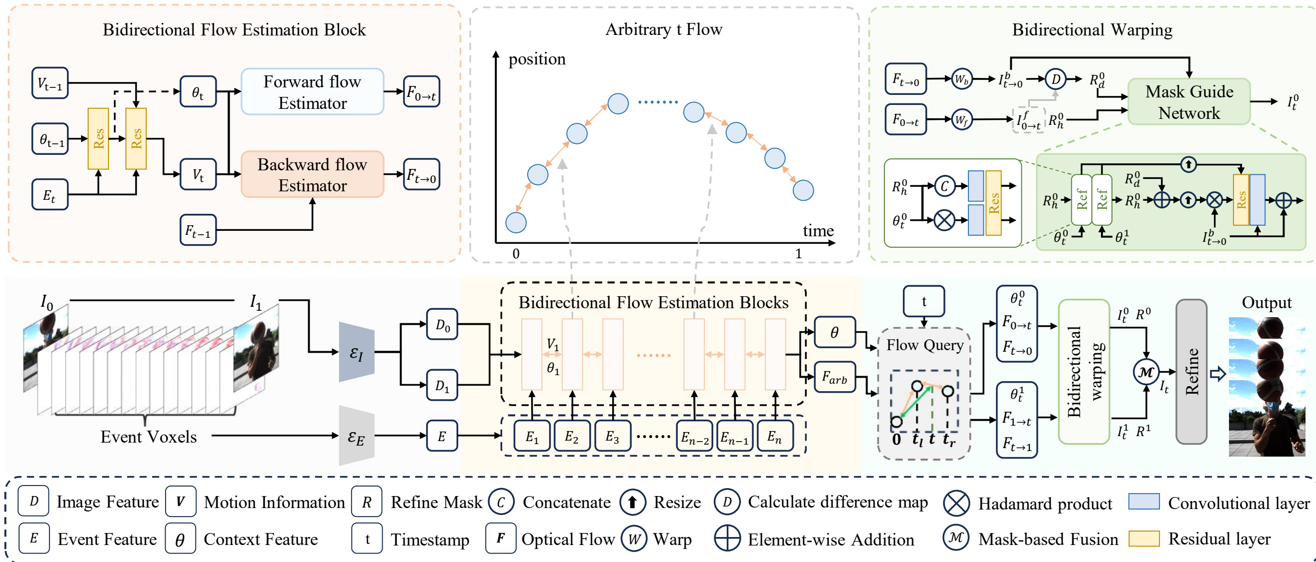

Figure 3. The overview of our method. The Event Feature is split by the channel dimension, and the Bidirectional Flow Estimation Block proposed shares weights.

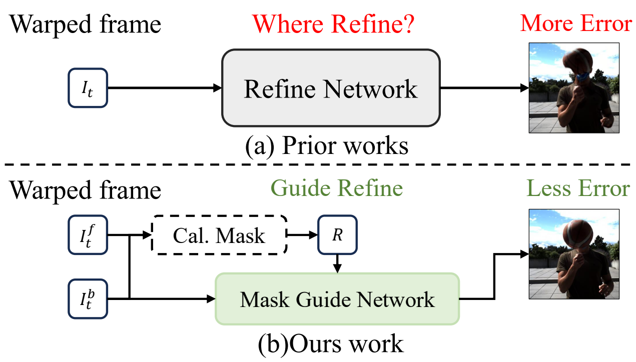

Figure 4. Visualization of the difference between our method and the previous method in the warping and refinement stages.

operation yields the final optical flow $F_{0\to t}$ and $F_{t\to 0}$, along with the interpolated contextual information θ_t^0.

3.4. Bidirectional Warping

Since the predicted optical flow is not perfectly accurate, the intermediate frames synthesized via image warping are usually of low quality. Consequently, prior works [24, 25, 44] typically feed these warped frames directly into a Refinement Network to generate a higher quality output, as illustrated in Fig. 4 (a). However, expecting the network to implicitly identify and correct the specific erroneous regions is challenging. Conversely, our proposed Bidirectional Warping (BiW) addresses this by calculating an explicit refine mask R to guide the network to focus on these areas that need repair, as shown in Fig. 4 (b).

Following the acquisition of the bidirectional optical flow and contextual information at time t via Flow Query, we leverage BiW to synthesize high-quality intermediate frames. The structure of BiW is presented in Fig. 3. We initially generate the low-quality intermediate frames I_t^f (forward-warped) and I_t^b (backward-warped) using the optical flow $F_{0\to t}$ and $F_{t\to 0}$. Subsequently, we compute the hole mask R_h^0 and the difference mask R_d^0. Refining the hole values in I_t^f poses a significant challenge. Instead, we opt to use R_h^0 and R_d^0 to correct errors in the backward warped image I_t^b to obtain the intermediate frame I_t. Since hole regions often coincide with occlusions, and backward warping is similarly prone to error in these regions, our chosen repair strategy is both reasonable and effective.

$$\begin{aligned} R_h^0 &= where(I_t^f = 0), \\ R_d^0 &= where(|I_t^f - I_t^b| > Y), \\ R^0, I_t^0 &= MG(R_h^0, R_d^0, \theta_t^0, \theta_t^1), \end{aligned} \tag{4}$$

where the function $where(\cdot)$ sets the pixel positions that meet the condition to 1. I_t^f and I_t^b are the images obtained via forward and backward warping. Y is a threshold. $MG(\cdot)$ denotes the mask guide network, which outputs the refined frame I_t^0 and the refine mask R^0. Specifically, we first employ R_h^0 to guide the Reference block (Ref) in extracting reference information from the corresponding regions of θ_t^0 and θ_t^1, and update R_h^0 because inaccuracies in optical flow may introduce errors in R_h^0. Subsequently, we add R_h^0 and R_d^0 to obtain refine mask R^0, multiply it by I_t^b to identify the pixels that need to be refined, and combine them with the reference information for the final prediction. It is noteworthy that R_d^0 is not fed into the Ref, as R_d^0 originates from the difference map between I_t^f and I_t^b. Since I_t^b is not necessarily incorrect in these regions, our objective is to direct the network's attention more toward these areas

Table 1. Regarding the evaluation on the GOPRO and SNU-FILM datasets. The best results are marked in **bold**, while the second are marked with underlines. For the GOPRO dataset, we interpolate all skipped frames in a single inference. An asterisk (*) denotes that our reproduction on the GoPro dataset failed to achieve their reported scores. Therefore, we adopt the reported superior performance from their original publication.

Method	Venue	Warping Type	GOPRO				SNU-FILM			
			Skip 7		Skip 15		hard		extreme	
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
RIFE [11]	ECCV'22	B	29.66	0.889	25.14	0.772	30.36	0.920	25.54	0.853
M2M [9]	CVPR'22	F	29.83	0.886	24.98	0.751	30.21	0.898	25.03	0.785
UPR-Net [14]	CVPR'23	F	27.91	0.855	24.61	0.758	30.82	0.928	25.61	0.862
IQ-VFI [8]	CVPR'24	F	-	-	-	-	30.83	0.938	25.45	0.863
BiM-VFI [34]	CVPR'25	B	-	-	-	-	30.30	0.897	24.87	0.780
Timelens [38]	CVPR'21	B	34.81	0.959	33.21	0.942	37.67	0.969	35.36	0.951
CBM-Net [18]	CVPR'23	B	36.07	<u>0.972</u>	35.46	0.966	36.52	<u>0.971</u>	35.07	0.958
TLXNet [29]	ECCV'24	B	37.06	0.970	36.43	<u>0.968</u>	37.67	<u>0.971</u>	36.10	<u>0.962</u>
TimeTracker* [25]	CVPR'25	B	<u>37.13</u>	0.962	<u>36.54</u>	0.958	<u>37.92</u>	0.967	<u>36.47</u>	0.959
Ours	-	F&B	**37.66**	**0.976**	**36.90**	**0.970**	**38.32**	**0.975**	**36.96**	**0.967**

Figure 5. Visual comparison among different methods on synthetic datasets.

rather than utilizing reference information for refine. Subsequently, the optical flow $F_{1\to t}$ and $F_{t\to 1}$ are processed by BiW to generate I_t^1 and the corresponding refine mask R^1 using the same operation.

Finally, we obtain the two intermediate frames I_t^0 and I_t^1 and their refine masks R^0 and R^1. Then, we use Eq. (5) to fuse them.

$$\begin{aligned} M &= softmax((1-R^0)*(1-t),(1-R^1)*t), \\ I_t &= I_t^0 * M^0 + I_t^1 * M^1. \end{aligned} \tag{5}$$

Obviously, when t is smaller, the intermediate frame is closer to keyframe I_0, so I_t^0 should be assigned a higher weight, I_t^1 should receive a lower weight. In addition, we use $1-R^0$ and $1-R^1$ to reduce the weight of regions that need to be refined, because we trust the regions that have no problems more. We use the softmax function to obtain the final weights M, which includes M^0 and M^1, which are the weights for I_t^0 and I_t^1 respectively. Then, we use these weights to obtain the weighted sum I_t. Finally, although our BiW can calculate the refine mask to guide the network to refine these regions, problems still exist. In regions where holes do not occur, if I_t^f and I_t^b appear with the same error, these error regions will not be calculated into the refine mask, thus generating erroneous results. Therefore, we use a Refine module like [11, 29] to further refine I_t to deal with this situation.

4. Experiment

4.1. Settings

Datasets. To evaluate our model against state-of-the-art methods, we use both synthetic and real-world datasets. Synthetic benchmarks include GOPRO [30] and SNU-FILM [5], where events are generated using V2E [10]. Real-world evaluation is conducted on BS-ERGB [39] and HS-ERGB [38], which feature challenging motion scenes.

Table 2. Regarding the evaluation on the BS-ERGB and HS-ERGB datasets. The best results are marked in **bold**, while the second are marked with underlines. An asterisk (*) denotes that our reproduction on the BS-ERGB dataset failed to achieve their reported scores. therefore, we adopt the reported superior performance from their original publication.

Method	Venue	Warping Type	BS-ERGB				HS-ERGB			
			Skip 1		Skip 3		Skip 5		Skip 7	
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
RIFE [11]	ECCV'22	B	25.62	0.765	23.44	0.728	32.62	0.857	31.15	0.936
M2M [9]	CVPR'22	F	25.53	0.779	23.24	0.740	32.07	0.861	30.61	0.837
UPR-Net [14]	CVPR'23	F	25.62	0.779	23.08	0.736	32.22	0.857	30.69	0.834
Timelens [38]	CVPR'21	B	27.16	0.783	25.85	0.765	32.76	0.861	31.87	0.851
CBM-Net [18]	CVPR'23	B	29.25	<u>0.814</u>	28.45	0.806	32.21	0.842	31.88	0.837
TLXNet [29]	ECCV'24	B	29.30	0.813	28.72	0.807	-	-	31.58	0.827
TimeTracker* [25]	CVPR'25	B	**29.85**	**0.823**	**29.14**	<u>0.807</u>	<u>33.59</u>	<u>0.872</u>	<u>32.68</u>	<u>0.861</u>
Ours	-	F&B	<u>29.76</u>	**0.823**	<u>29.03</u>	**0.815**	**34.63**	**0.889**	**34.19**	**0.881**

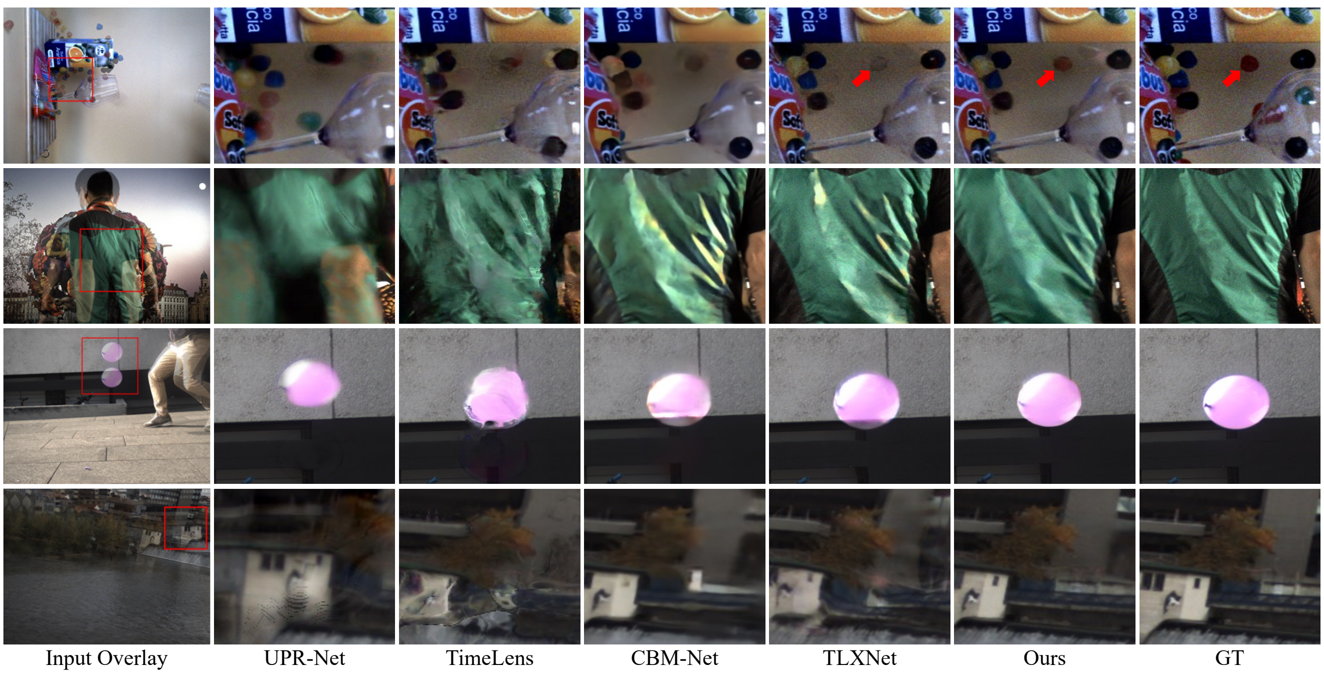

Figure 6. Visual comparison among different methods on real datasets.

Training Settings. We trained our model end-to-end on the GOPRO dataset using L1 and LPIPS [45] losses. Training is performed for 20 epochs with Adam [19], starting at a learning rate of 10^{-4} and decayed to 10^{-6} via cosine annealing. Images and events are randomly cropped to 256×256. For BiFEB, we set $n = 16$ to ensure small time intervals. All experiments are conducted on a single NVIDIA RTX 3090 GPU. *Specific training details are presented in the supplementary material.*

State-of-the-Art Methods (SOTA). To validate our method, we compare it against representative SOTA flow-based methods: (1) Forward Warping-based VFI Methods: M2M-PWC [9], UPR-Net [14], IQ-VFI [8]. (2) Backward Warping-based VFI Methods: RIFE [11], BiM-VFI [34]. (3) Backward Warping-based E-VFI: Timelens [38], CBM-Net [18], TLXNet [29], TimeTracker [25]. *The comparison with synthesis methods is included in the supplementary material.* For a fair comparison, we followed their publicly released code to retrain these methods under the same conditions and performed fine-tuning to get more ac-

Table 3. Analysis of computational cost on the GOPRO Dataset. The best performance in each column is marked in **bold**, and the second best is marked with underlines. The measured costs include Memory (peak GPU memory usage), MACs/f (average MACs per interpolated frame), and Time/f (average time per interpolated frame). OOM means Out of GPU Memory (>24GB)

Method	31 frames			63 frames			127 frames		
	Memory	MACs/f	Time/f	Memory	MACs/f	Time/f	Memory	MACs/f	Time/f
Timelens [38]	**1.93GB**	1535.28G	1.065s	**1.93GB**	1535.28	1.031s	**1.93GB**	1535.28G	1.126s
CBM-Net [18]	10.76GB	3732.99G	2.652s	10.77GB	3732.99G	2.589s	10.77GB	3732.99G	2.959s
TLXNet [29]	11.70GB	**729.45G**	**0.079s**	OOM	-	-	OOM	-	-
Ours	5.29GB	887.09G	0.137s	5.95GB	**738.09G**	**0.117s**	7.27GB	**665.35G**	**0.108s**

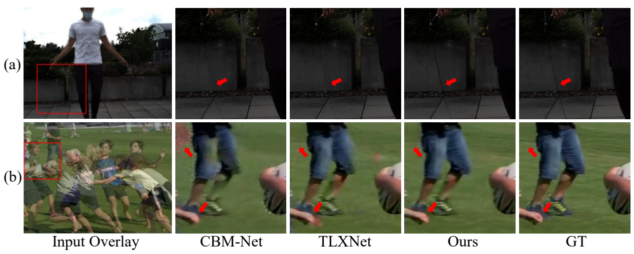

Figure 7. Visual comparison of methods under extreme motion

curate results. Since TimeTracker is not open-source, our replicated implementation failed to reproduce their results. Consequently, to ensure fair comparison, we adopt the reported superior performance from their original publication. *The production results of TimeTracker will be placed in the supplementary material.* We use the SSIM and PSNR metrics to evaluate the performance of all models.

4.2. Evaluation on Synthesis Dataset

Quantitative Comparison. As shown in Tab. 1, our method consistently outperforms all compared methods on both the GoPro and SNU-FILM datasets. This superior performance demonstrates two key aspects: first, our proposed BiFEB module is better equipped to handle complex motion in challenging scenes; and secondly, our BiW module can leverage bidirectional optical flow to synthesize more accurate intermediate frames.

Qualitative Comparison. Fig. 5 presents a detailed qualitative comparison of our approach against several state-of-the-art methods. We observe that UPR-Net and Time-Lens frequently fail to accurately estimate complex motion, which often results in severe motion blurring. While CBM-Net and TLXNet achieve better motion prediction, they still suffer from deformation and color difference. In contrast, our method achieves the best performance. This is primarily because our proposed BiFEB module provides more accurate optical flow estimation. Concurrently, the BiW module plays a crucial role by effectively guiding the network to restore and refine challenging areas, particularly along object boundaries, which are highly susceptible to warping errors, thereby yielding the highest quality output. *Additional visualization results are presented in the supplementary material.*

4.3. Evaluation on Real Dataset

Quantitative Comparison. As shown in Tab. 2, our method achieves consistently strong performance on both BS-ERGB and HS-ERGB. On BS-ERGB, it ranks within the top two across all skipping levels. On HS-ERGB, our method surpasses all competitors and obtains the highest PSNR and SSIM in every setting, with especially notable gains under challenging motion. These results demonstrate the robustness of our bidirectional warping strategy across varying motion intensities.

Qualitative Comparison. Fig. 6 demonstrates the qualitative comparison results on the real-world dataset. We observe that other methods frequently suffer from noticeable color shift artifacts in complex scenes, highlighted by the examples of the candy, jacket, and basketball. In contrast, our method consistently yields the highest visual fidelity. Compared to other methods that perform direct refinement, our BiW actively guides the network to restore and correct these erroneous regions. Furthermore, thanks to BiFEB providing accurate optical flow, we also predict object motion more accurately. *Additional visualization results are presented in the supplementary material.*

4.4. Computational Cost Analysis

To compare the computational cost of our method against other approaches, we evaluate the resources required for interpolating varying numbers of frames on the GOPRO dataset. The results are summarized in Tab. 3. As Time-Lens and CBM-Net are backward methods, they necessitate performing optical flow estimation and interpolation for every single output frame. Consequently, their per-frame cost remains nearly constant (minor variations may result from hardware fluctuations). Conversely, our method's total cost amortizes over the number of frames because we only require a single optical flow estimation; thus, the per-frame computational cost decreases significantly when interpolat-

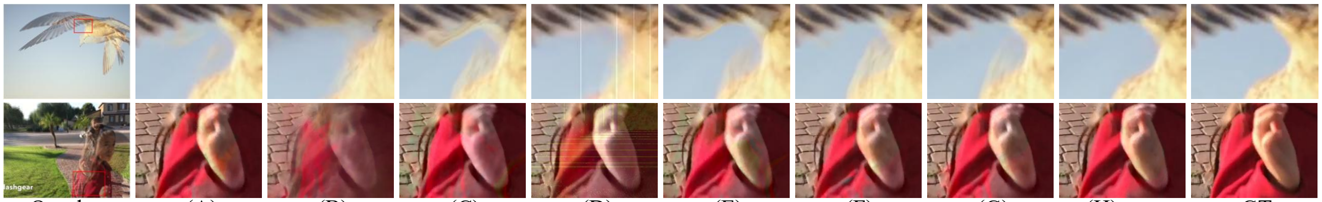

Figure 8. Visual comparison under different settings.

Table 4. Ablation studies on BiFEB and BiW.

Variants	Flow Estimator	Interpolation	PSNR	SSIM
A	RAFT+Timelens	BiW	36.27	0.962
B	BiFEB (n=4)	BiW	29.75	0.892
C	BiFEB (n=8)	BiW	35.24	0.954
D	BiFEB (n=16)	Forward	30.54	0.912
E	BiFEB (n=16)	Backward	35.78	0.959
F	BiFEB (n=16)	BiW (w/o R_h)	36.01	0.961
G	BiFEB (n=16)	BiW (w/o R_d)	36.74	0.966
H(Ours)	BiFEB (n=16)	BiW	**36.96**	**0.967**

ing multiple frames. It is worth noting that TLXNet has lower MACs/f and Time/f when interpolating 31 frames, because they use a large amount of GPU memory to store intermediate variables to increase speed. In comparison, our method has three advantages: (1) As verified in Sec. 4.2 and Sec. 4.3, we have better interpolation quality. (2) When interpolating a large number of frames, TLXNet's strategy of using GPU memory to increase speed leads to OOM, while we have lower MACs/f and the interpolation speed is also very close. (3) TLXNet cannot interpolate frames at arbitrary time instances (e.g., $t = 0.51, t = 0.88$), while we can infinitely interpolate frames at arbitrary time instances. Therefore, we believe that our method has better performance compared to TLXNet.

4.5. Evaluation on Extreme Motion

To specifically validate the enhanced motion estimation capabilities of our proposed model, we conducted a qualitative comparison against competing methods under challenging, extreme motion scenarios. The results are shown in Fig. 7.

Scenario a (Thin Structures): This scene features a very thin rope moving at high velocity. Other methods struggle with this combination, often resulting in broken or blurred structures. Our method, however, accurately predicts the rope's trajectory and position in the interpolated frame, preserving its integrity.

Scenario b (Occlusion): This example involves complex human motion. CBM-Net suffers from severe artifacts, erroneously predicting the appearance of a person before they are present in the ground truth. Similarly, TLXNet fails to accurately reconstruct the moving hand. Our method successfully captures these intricate dynamics, precisely estimating the position of the hand and avoiding predictive errors.

This superior motion estimation is a direct result of BiFEB. BiFEB effectively decomposes the object's complex movement into multiple simple motions in the temporal domain, allowing for a refined prediction of the underlying motion trajectory. This decomposition yields a significantly more accurate optical flow field, which is critical for synthesizing fine details and predicting high-speed objects.

4.6. Ablation Study

We perform ablation studies on the two core modules of our model, BiFEB and BiW. Results are shown in Tab. 4. Variant A uses RAFT to predict forward optical flow and Timelens to predict backward optical flow, where performance degradation is caused by errors in linear optical flow. Variants B and C show that as n increases, BiFEB predicts nonlinear flow more accurately. Results from Variants D–G indicate that our BiW module guides the network to rectify erroneous regions, leading to better outcomes. Fig. 8 presents partial visualizations of different variants. It shows that our method predicts object motion more accurately than others, rectifies erroneous regions, and achieves smaller color discrepancies, further demonstrating the effectiveness of the BiFEB and BiW modules.

5. Conclusion

In this paper, we introduced "One-Shot Flow, Any-Time Frame," a novel framework that fundamentally addresses the trade-off between computational efficiency and synthesis quality in event-based video frame interpolation. Our core innovation is the Bidirectional Flow Estimation Block (BiFEB), which computes a comprehensive motion representation in a single pass. This allows our Flow Query (FQ) module to instantly retrieve optical flows for an arbitrary timestamp, making the framework highly efficient and scalable. Complemented by a sophisticated Bidirectional Warping (BiW) module that explicitly repairs artifacts, our method generates high-quality frames. Comprehensive experiments validate that our approach consistently surpasses existing methods in both quality and efficiency.

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